

tcn_lite (full causal TCN) - selected family

Full-convolution causal TCN, 24 filters: the selected model tcn_lite_ft.

Selected: tcn_lite_ft_int8. Same graph as tcn_lite, fine-tuned for a few more epochs. It loses to no other candidate in the int8 closed loop on validation. Final test (10 subjects): open loop 28.59 ms vs rls_ar 31.15; closed loop 28.25 / 30.20 / 33.07 / 41.55 ms at 10/30/50/80% dropped; no invented or displaced RSA peak.

Architecture

- Pointwise Conv1D stem (24 filters) + BatchNorm + ReLU, then 4 residual blocks of causal full Conv1D (kernel 3, dilations 1, 2, 4, 7) with explicit left padding, BatchNorm (folded at export) and ReLU.
- Receptive field 29 beats (window model = streaming model). The newest time step is concatenated with the 9 features, then Dense(32, ReLU) - Dense(1).
- 8561 parameters, 8048 MACs per beat in streaming mode (under the 10k limit), 8.4 KB int8; int8 costs +0.5% on test.

Layer table of tcn_lite (B = batch)

layer	configuration	output shape	params
InputLayer		(B, 30, 2)	0
Conv1D	filters=24	(B, 30, 24)	48
BatchNormalization		(B, 30, 24)	96
ReLU		(B, 30, 24)	0
ZeroPadding1D	padding=(2, 0), left=2	(B, 32, 24)	0
Conv1D	filters=24, kernel_size=3	(B, 30, 24)	1728
BatchNormalization		(B, 30, 24)	96
ReLU		(B, 30, 24)	0
Add		(B, 30, 24)	0
ZeroPadding1D	padding=(4, 0), left=4	(B, 34, 24)	0
Conv1D	filters=24, kernel_size=3, dilation_rate=2	(B, 30, 24)	1728
BatchNormalization		(B, 30, 24)	96
ReLU		(B, 30, 24)	0
Add		(B, 30, 24)	0
ZeroPadding1D	padding=(8, 0), left=8	(B, 38, 24)	0
Conv1D	filters=24, kernel_size=3, dilation_rate=4	(B, 30, 24)	1728
BatchNormalization		(B, 30, 24)	96
ReLU		(B, 30, 24)	0
Add		(B, 30, 24)	0
ZeroPadding1D	padding=(14, 0), left=14	(B, 44, 24)	0
Conv1D	filters=24, kernel_size=3, dilation_rate=7	(B, 30, 24)	1728
BatchNormalization		(B, 30, 24)	96
ReLU		(B, 30, 24)	0
Add		(B, 30, 24)	0
Cropping1D		(B, 1, 24)	0
Flatten		(B, 24)	0
InputLayer		(B, 9)	0
Concatenate		(B, 33)	0
Dense	units=32, activation=relu	(B, 32)	1088
Dense	units=1	(B, 1)	33

Inputs and feature engineering (tcn_lite)

- Sequence input** (TCN/GRU): the last 30 RR intervals as two channels, z-RR and z-dRR (first step dRR = 0), z-scored with mean/std fitted on the training subjects only (zero variance -> std 1).
- Tabular input:** the engineered features of the newest beat (see 01_feature_engineering.pdf), z-scored per feature with training statistics.
- Target:** $dRR[n+1] = RR[n+1] - RR[n]$. Residual models predict the z-scored correction over the RLS forecast (dRR - rls_pred); the output is de-normalised and added back to rls_pred.

Tabular features (9):

feature	definition
rmssd	root mean square of the 29 successive differences
rs	rescaled range: (max - min of the cumulative sum of RR - mean) / std
ccm	complex correlation measure: Poincare-triangle area / (pi SD1 SD2 (W-3)), clipped to [0, 1]
guzik	Guzik asymmetry: sum of positive dRR / sum of dRR
rls_pred	RLS AR(8) forecast of the next dRR (ms), clamped to +/-300
rls_err	last RLS innovation: true dRR minus the previous forecast (ms)
bp	RSA band-pass output: RBJ biquad on raw RR, centre 0.30 cycles/beat, Q 1
bp_prev	previous band-pass output (gives the phase with bp)
run_len	signed length of the current run of same-sign dRR (+ = decelerating)

Training

Keras, AdamW (weight decay 1e-4), Huber loss (delta 1, z units), batch 512, steps_per_execution 64, epochs of 2M random training windows (about 10% of the 20.8M), EarlyStopping on val_loss (restore best weights) and ReduceLROnPlateau (factor 0.5, min 1e-5), seed 211. int8: full-integer post-training quantisation (TFLite, int8 in/out), deterministic best-of-K calibration (8 draws of 2k/5k training windows; the draw with the smallest gap on held-out training subjects is kept).

Variants trained

run	change vs parent	feature set (n)	params	MACs/beat	training	epochs
tcn_lite	round 1 base (about 48 full passes)	seq (9)	8561	8048	from scratch, lr 0.001	48
tcn_lite_ft	same model, fine-tuned (lr 3e-4); control for the DAD run. SELECTED	seq (9)	8561	8048	fine-tune of tcn_lite, lr 0.0003	7
tcn_lite_dad	fine-tuned on closed-loop-filled data (DAD)	seq (9)	8561	8048	fine-tune of tcn_lite, lr 0.0003, closed-loop data share 0.5	18
tcn_lite_rsa	+ gated RSA features (seq_rsa), from scratch	seq_rsa (13)	8689	8176	from scratch, lr 0.001	37
tcn_lite_rsa_ws	+ gated RSA features, warm start (new inputs zero-initialised)	seq_rsa (13)	8689	8176	fine-tune of tcn_lite, lr 0.0003	7

Results

model	val open loop	closed-loop model	10%	30%	50%	80%	RSA invented at 80%	test whole-series 80%	val source
tcn_lite	32.64	int8	25.91	29.20	33.23	41.82	4%	43.73	round 4
tcn_lite_ft	32.62	int8	25.87	29.12	33.13	41.66	0%	43.74	round 4
tcn_lite_dad	33.09	int8	26.32	29.34	33.22	41.38	4%	42.91	round 4
tcn_lite_rsa	33.06	int8	26.18	29.40	33.23	42.30	0%	45.97	round 4
tcn_lite_rsa_ws	32.62	int8	25.87	29.10	33.12	41.86	0%	43.97	round 4

RMSE in ms. **val open loop**: one-step-ahead on the true history, 16 validation subjects. **10-80%**: closed loop (pacemaker use) on validation: that share of beats dropped at random from beat 0, each filled by the model and fed back; RMSE on model-filled beats, first 5000 beats per subject, mean of the seeds of that round (3 seeds from round 2 on). int8 is used where it exists. **RSA invented**: share of subject-runs whose filled series gains an RSA peak the original lacks. **test whole-series 80%**: diagnostic on the 10 test subjects (whole series, seed 7, results/test/ folder); it was not used for any selection.

Correlation between the original and the filled series (validation and test subjects)

model	corr RR validation	corr RR test	corr dRR validation	corr dRR test	RMSE filled, validation (ms)	RMSE filled, test (ms)
tcn_lite	0.9352	0.9394	0.270	0.264	50.27	43.73
tcn_lite_ft	0.9356	0.9395	0.272	0.266	50.15	43.62
tcn_lite_dad	0.9383	0.9425	0.282	0.278	49.26	42.91
tcn_lite_rsa	0.9311	0.9337	0.261	0.246	51.87	45.97
tcn_lite_rsa_ws	0.9350	0.9385	0.270	0.260	50.40	43.97
tcn_lite_ft_int8	0.9354	0.9393	0.269	0.263	50.18	43.74

Computed on the 16 validation and the 10 test subjects (results/validation/ and results/test/ folders), on the ENTIRE series, after dropping 80% of the beats (seed 7, from beat 0) and filling them in closed loop with the model (Burg-AR fill for the first 30 beats): **corr RR** = `np.corrcoef(original_series, modified_series)[0, 1]`; **corr dRR** = the same function on the beat-to-beat changes, `np.corrcoef(np.diff(original), np.diff(filled))[0, 1]`, which isolates the short-term dynamics. Mean over subjects; per-subject values in <folder>/crosscorr.csv. **RMSE filled**: RMSE on the model-filled beats of the same runs. Diagnostic: the selection used the round-4 protocol, not these runs.

Per validation subject (16): deployed model vs TCN_MHA and rls_ar

subject	RR tcn_lite_ft_int8	RR tcn_mha_current	RR rls_ar	dRR tcn_lite_ft_int8	dRR tcn_mha_current	dRR rls_ar
000	0.9377	0.9021	0.9286	0.305	0.161	0.248
006	0.9225	0.8703	0.9068	0.220	0.096	0.165
008	0.9178	0.8873	0.9056	0.245	0.131	0.204
16273	0.8988	0.8366	0.8888	0.288	0.129	0.262
16483	0.9453	0.9423	0.9336	0.258	0.233	0.186
18177	0.9392	0.9378	0.9278	0.255	0.242	0.203
19088	0.9097	0.9007	0.9072	0.242	0.191	0.239
nsr010RRcl	0.9532	0.9421	0.9364	0.300	0.217	0.182
nsr016RRcl	0.9758	0.9711	0.9678	0.256	0.169	0.177
nsr018RRcl	0.9309	0.9173	0.8982	0.412	0.300	0.250
nsr022RRcl	0.9171	0.8718	0.9063	0.275	0.155	0.236
nsr038RRcl	0.9455	0.9445	0.9342	0.158	0.151	0.108
nsr041RRcl	0.9467	0.9446	0.8636	0.274	0.259	0.140
nsr045RRcl	0.9455	0.9344	0.9340	0.365	0.266	0.264
nsr048RRcl	0.9288	0.9019	0.9174	0.221	0.125	0.174
nsr054RRcl	0.9521	0.9477	0.9419	0.233	0.192	0.192

Per test subject (10): deployed model vs TCN_MHA and rls_ar

subject	RR tcn_lite_ft_int8	RR tcn_mha_current	RR rls_ar	dRR tcn_lite_ft_int8	dRR tcn_mha_current	dRR rls_ar
003	0.9053	0.9002	0.8973	0.289	0.263	0.263
007	0.9464	0.9399	0.9410	0.229	0.191	0.200
16420	0.9360	0.9172	0.9087	0.252	0.138	0.226
16539	0.8188	0.7619	0.8066	0.338	0.183	0.331
16786	0.9341	0.9009	0.9276	0.279	0.115	0.257
nsr003RRcl	0.9453	0.9368	0.9298	0.297	0.220	0.261
nsr013RRcl	0.9816	0.9816	0.9795	0.247	0.246	0.201
nsr026RRcl	0.9891	0.9825	0.9774	0.236	0.128	0.114
nsr033RRcl	0.9567	0.9190	0.9345	0.223	0.099	0.154
nsr034RRcl	0.9796	0.9778	0.9734	0.238	0.209	0.178

Verdict

- tcn_lite_ft_int8 beats micro_tcn at 10% and 30% dropped ($p \leq 0.003$, float agrees) and tcn_lite pooled; it ties only tcn_lite_dad (same size; DAD's 0.28 ms gain at 80% is not significant).
- The gated RSA features add nothing at 10-50% and cost 0.2 ms at 80%; tcn_lite_rsa from scratch is confounded by a 13x smaller training budget.

- At 80% dropped every model damps RSA power (about $\times 0.5$): an information limit, not a defect; the selected model invented no RSA peak on validation or test; its displaced-peak flags on validation (only the 2 RSA subjects) were no more frequent than rls_ar's, and there were none on test.

Abbreviations and acronyms used in this document

term	meaning
AdamW	Adam optimiser with decoupled weight decay (the training algorithm of the neural models)
AR	autoregressive: a value predicted as a weighted sum of past values; AR(p) uses p past values
ar	feature-set name: autoregressive inputs (rr_dev and the 8 most recent dRR) of the linear model
B	batch dimension (number of windows processed together) in the layer output shapes
bp	feature: RSA band-pass output: RBJ biquad on raw RR, centre 0.30 cycles/beat, Q 1
bp_prev	feature: previous band-pass output (gives the phase with bp)
Burg-AR	AR model whose coefficients were fitted with Burg's method (used for the first 30 beats)
ccm	complex correlation measure: non-linearity index of the Poincare plot of consecutive RR intervals
Conv1D	one-dimensional convolution layer
DAD	data as demonstrator: training on inputs that contain the model's own closed-loop fills, with the true target (a remedy for exposure bias)
_dad	run-name suffix: fine-tuned with DAD (closed-loop-filled) training data
dRR	beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted
_ft	run-name suffix: fine-tuned (training continued from the parent's weights)
GRU	gated recurrent unit (a recurrent neural-network cell)
guzik	feature: Guzik asymmetry: sum of positive dRR / sum of dRR
HRV	heart rate variability
int8	8-bit integer: weights and activations quantised to 8-bit integers, as run on the microcontroller; _int8 = the int8 version of a run
KB	kilobyte (1,024 bytes)
lr	learning rate (step size of the optimiser)
MAC	multiply-accumulate operation (one multiplication plus one addition): the unit of compute cost on the microcontroller; MACs/beat = how many the model needs per heartbeat
MACs	multiply-accumulate operations (see MAC)
MHA	multi-head attention (a transformer layer, not supported by the MCU toolchain)
ms	milliseconds
np.corrcoef	NumPy's Pearson correlation-coefficient function
np.diff	NumPy's successive-difference function
nsrNNRRcl	subject code of an RR series from the normal sinus rhythm (NSR) recordings (file data/series/<code>.txt); purely numeric codes such as 16273 or 006 are other subjects
p	p-value of the paired per-subject Wilcoxon test: probability of a difference at least this large if the two models were equally accurate ($p < 0.05$ = significant); in AR(p), the number of past values
Q	quality factor of a band-pass filter: centre frequency / bandwidth
RBJ	Robert Bristow-Johnson's standard biquad (second-order) filter formulas
ReLU	rectified linear unit, $\max(0, x)$ (an activation function)
RLS	recursive least squares: an adaptive filter that updates its coefficients at every beat
rls_ar	baseline: adaptive per-patient AR(8) forecast by RLS
rls_err	feature: last RLS innovation: true dRR minus the previous forecast (ms)
rls_pred	feature: RLS AR(8) forecast of the next dRR (ms), clamped to ± 300
RMSE	root mean square error (ms): the main accuracy measure, lower is better
rmssd	root mean square of successive differences of the RR intervals in the window
RR	RR interval: time between two consecutive heartbeats (R waves of the ECG), in ms
rs	rescaled range: $(\max - \min \text{ of the cumulative deviations from the mean}) / \text{standard deviation}$
RSA	respiratory sinus arrhythmia: the oscillation of the heart rate driven by breathing
_rsa	run-name suffix: uses the RSA-tracker features
run_len	feature: signed length of the current run of same-sign dRR (+ = decelerating)
SD1	Poincare-plot standard deviation across the identity line (short-term variability)
SD2	Poincare-plot standard deviation along the identity line (long-term variability)
seq	feature-set / model-kind name: sequence model (takes the raw RR window plus tabular features)
TCN	temporal convolutional network (causal, dilated 1-D convolutions)
tcn	temporal convolutional network (run-name part)
TFLite	TensorFlow Lite: the runtime and file format for models on embedded devices
val	validation: the 16 validation subjects, used for model selection
W	window length: the 30 most recent RR intervals
_ws	run-name suffix: warm start (initialised from the parent's weights, new inputs set to zero)

term	meaning
z	z-score: (value - mean) / standard deviation, with statistics from the training subjects